

Notes: Halac and Yared (JPE, 2025)

A Theory of Fiscal Responsibility and Irresponsibility

Contents

1	Big picture	3
2	Model setup and key objects	3
2.1	Environment, players, and timing	3
2.2	Social welfare and government welfare	4
2.3	Information, discretion, and debt bounds	4
2.4	Histories, strategies, and continuation values	5
3	Models and theoretical results (all equations + interpretation)	5
3.1	Equilibrium constraints: private information and limited commitment	5
3.2	Rewriting welfare: local IC and the $Q(v)$ object	6
3.3	Benchmarks: what happens without one of the frictions?	6
3.4	Proposition 1: bang-bang continuation values	7
3.5	Proposition 2: the fiscally responsible regime	7
3.6	Proposition 3: the fiscally irresponsible regime	8
3.7	Why the bad regime is temporary	8
3.8	Regime dynamics	8
3.9	Analytical example: log utility and saving rates	9
3.10	Factorization algorithm and proposition 4	10
3.11	Numerical simulation	10
4	Mechanism and assumptions (what makes the result work)	11
4.1	Why both frictions are necessary	11
4.2	Why the responsible regime is a deficit limit	11
4.3	Why the irresponsible regime is a surplus limit	11
4.4	Why crises trigger regime transitions	11

4.5	Why a larger bias makes regimes possible	11
4.6	Relation to the literature	12
5	Figures and proposition map	12
5.1	Figure 1: empirical motivation	12
5.2	Proposition 1 and Figure 2: the two-regime architecture	12
5.3	Figure 3: the fixed-point logic of regime existence	13
5.4	Figure 4: numerical simulation of regime transitions	14
5.5	Proposition map: how the results fit together	14
6	Short summary	14
7	Conclusion	15
7.1	Core conclusion	15
7.2	Economic intuition	15
7.3	Takeaway	15

1 Big picture

- **Question.** Why do countries move between long periods of fiscal responsibility and fiscal irresponsibility, and why do transitions in *both* directions tend to occur during crises?
- **Setting.** The paper studies an infinite-horizon small open economy with successive governments that choose borrowing and spending. Each government is deficit biased and privately observes the current fiscal need shock.
- **Core challenge.** A government wants to overspend today, but any punishment or reward must be delivered by *future* governments. Since current fiscal needs are privately observed, future governments cannot condition directly on past shocks; they can only react to observed fiscal choices.
- **Main theoretical result.** The best equilibrium for society is *bang-bang*: on the equilibrium path, continuation welfare is always either the highest feasible value or the lowest feasible value. This generates two endogenous fiscal regimes.
- **Responsible regime.** Fiscal responsibility takes the form of a *maximally enforced deficit limit*. Governments that keep borrowing below the limit are rewarded with continuation in the good regime; governments that break it are sent to the bad regime.
- **Irresponsible regime.** Fiscal irresponsibility takes the mirror form of a *maximally enforced surplus limit*. Governments that borrow enough are rewarded with a return to the good regime; governments that fail to do so remain in the bad regime.
- **Regime dynamics.** Transitions are triggered by sufficiently large fiscal shocks. In the responsible regime, severe shocks make governments willing to break the deficit limit. In the irresponsible regime, severe shocks make governments willing to choose large enough deficits to earn a return to responsibility.
- **Key comparative static.** Fiscal regimes can arise only if governments' deficit bias is large enough and the discount factor is high enough. If the bias is too small, the only sustainable outcome is the Markov one with no regime switching.
- **Economic takeaway.** The same political distortion that tends to create debt accumulation can also sustain temporary fiscal discipline, because the threat of future irresponsibility becomes severe enough to deter current overborrowing.

2 Model setup and key objects

2.1 Environment, players, and timing

- **Horizon and political turnover.** Time is infinite, $t \in \{0, 1, 2, \dots\}$, and a different government is in office each period.
- **Shock.** At the start of each period, an i.i.d. fiscal shock $v_t > 0$ is drawn from a bounded support $\Theta = [\underline{v}, \bar{v}]$ with density $f(v) > 0$.
- **Interpretation of the shock.** A large v_t means that the social value of deficit-financed spending is high — for example, because the economy is in a crisis or recession.

- **Choices.** The government chooses debt b_t and spending g_t .
- **Budget constraint.**

$$g_t = \tau - Rb_{t-1} + b_t,$$

where $\tau > 0$ is exogenous tax revenue and $R > 1$ is the gross interest rate on government bonds.

2.2 Social welfare and government welfare

- **Society's objective.**

$$V_t = \mathbb{E}_t \left[\sum_{k=0}^{\infty} \delta^k v_{t+k} U(g_{t+k}) \right], \quad V_t = \mathbb{E}_t [v_t U(g_t) + \delta V_{t+1}],$$

where $\delta \in (0, 1)$ and $U(\cdot)$ is strictly increasing and strictly concave.

- **Government's objective.**

$$\alpha v_t U(g_t) + \delta V_{t+1}, \quad \alpha > 1.$$

- **Deficit bias.** The parameter α captures the idea that the incumbent overvalues current spending relative to society — for example, because it can direct spending toward favored constituencies or categories.
- **Important asymmetry.** A government is biased only while in office. From next period onward, it values future welfare the same way society does. This asymmetry is what makes future punishments and rewards meaningful.

2.3 Information, discretion, and debt bounds

- **Private information.** The current government privately observes v_t . Future governments observe past debt choices, but not the exact past fiscal need shocks.
- **Interpretations.**
 - The incumbent may know the current cost of public goods better than future governments.
 - The incumbent may better observe current aggregate preferences or spending needs.
 - More generally, fiscal needs are harder to verify than fiscal choices such as deficits and debt.
- **Discretion.** Governments have full discretion over debt subject only to feasibility. There is no exogenously enforced fiscal rule.
- **Debt bounds.** Debt must satisfy

$$b_t \in [\underline{b}(b_{t-1}), \bar{b}(b_{t-1})],$$

with bounds chosen so that payoffs remain finite. Economically, these bounds rule out explosive Ponzi-type paths.

- **Key implication.** Since no outside authority enforces discipline, the only way to sustain good behavior is through self-enforcing continuation play by future governments.

2.4 Histories, strategies, and continuation values

- **History.** Let

$$h_{t-1} = \{b_{-1}, b_0, \dots, b_{t-1}\}$$

denote the history of debt choices up to period $t - 1$.

- **Resources.** Given inherited debt, available resources are

$$q_t(h_{t-1}) = \tau - Rb_{t-1}.$$

Thus spending can be written as $g_t = q_t(h_{t-1}) + b_t$.

- **Strategy.** A period- t government's strategy is $j_t(h_{t-1}, v_t)$, specifying debt as a function of history and type.
- **Continuation value to society.**

$$V_t(h_{t-1}) = \mathbb{E}_t[v_t U(q_t + b_t) + \delta V_{t+1}(h_t)].$$

- **Best and worst continuation equilibria.** The paper defines

$$\bar{V}(b) \quad \text{and} \quad \underline{V}(b)$$

as the highest and lowest continuation values for society that can be sustained after a given debt level b .

- **Best equilibrium for society.** Among all equilibria, the paper studies the one that maximizes date-0 social welfare.

3 Models and theoretical results (all equations + interpretation)

3.1 Equilibrium constraints: private information and limited commitment

Given resources q associated with inherited debt, the recursive problem assigns each type v a debt level $b(v)$ and a continuation value $V(b(v))$.

The **private information constraint** requires truthful behavior across types:

$$\alpha v U(q + b(v)) + \delta V(b(v)) \geq \alpha v U(q + b(v')) + \delta V(b(v')) \quad \text{for all } v, v' \in \Theta.$$

The **limited commitment constraint** requires that the government not deviate publicly to some off-schedule debt level b' :

$$\alpha v U(q + b(v)) + \delta V(b(v)) \geq \alpha v U(q + b') + \delta V(b').$$

To summarize the government's best off-schedule deviation under the *worst* continuation value, the paper defines

$$b^p(q, v) \in \arg \max_{b \in [\underline{b}, \bar{b}]} \{ \alpha v U(q + b) + \delta \underline{V}(b) \}.$$

This is the government's flexible borrowing choice when it is punished as harshly as possible.

The recursive **best-equilibrium program** is

$$\bar{V}(b_{-1}) = \max_{(b(v), V(b(v)))} \mathbb{E}[vU(q + b(v)) + \delta V(b(v))]$$

subject to the private information constraint, the limited commitment constraint, and feasibility:

$$b(v) \in [\underline{b}(b_{-1}), \bar{b}(b_{-1})], \quad V(b(v)) \in [\underline{V}(b(v)), \bar{V}(b(v))].$$

The symmetric minimization problem defines the worst feasible continuation value:

$$\underline{V}(b_{-1}) = \min_{(b(v), V(b(v)))} \mathbb{E}[vU(q + b(v)) + \delta V(b(v))].$$

Interpretation. The planner is not choosing policy directly. Instead, it is choosing a *self-enforcing schedule* of borrowing and continuation values. Truth telling and obedience must both be credible.

3.2 Rewriting welfare: local IC and the $Q(v)$ object

Using the envelope logic from local incentive compatibility, government welfare for type v can be written as

$$\alpha v U(q + b(v)) + \delta V(b(v)) = \alpha \underline{v} U(q + b(\underline{v})) + \delta V(b(\underline{v})) + \int_{\underline{v}}^v \alpha U(q + b(\tilde{v})) d\tilde{v}.$$

Substituting this into the objective yields a convenient social-welfare representation:

$$\alpha \underline{v} U(q + b(\underline{v})) + \delta V(b(\underline{v})) + \alpha \int_{\underline{v}}^{\bar{v}} U(q + b(v)) Q(v) dv,$$

where

$$Q(v) = 1 - F(v) - v f(v) \left(1 - \frac{1}{\alpha}\right).$$

- The term $1 - F(v)$ is the usual *virtual-surplus* force from incentive problems: raising borrowing for type v also raises rents for all higher types.
- The term $-v f(v)(1 - 1/\alpha)$ is the political-economy wedge: because $\alpha > 1$, society values current borrowing less than the incumbent government does.
- The paper imposes an assumption that there exists $\hat{v}$ such that $Q'(v) < 0$ for $v < \hat{v}$ and $Q'(v) > 0$ for $v > \hat{v}$. This is what pins down the shape of the optimal distortion.

Interpretation. $Q(v)$ tells us where society wants to place distortions in the borrowing schedule once both information rents and deficit bias are taken into account.

3.3 Benchmarks: what happens without one of the frictions?

- **No deficit bias** ($\alpha = 1$). Then governments and society agree on the value of current spending. The best equilibrium implements the first-best allocation even if shocks are private. There is no need for fiscal regimes.

- **Public shocks.** If v is observable and $\alpha > 1$, only limited commitment matters. Public deviations can be punished directly off path, and the best equilibrium simply restarts each period at the best feasible continuation value.
- **Main lesson.** Neither deficit bias alone nor private information alone generates regime dynamics. The paper's result needs the *combination* of the two frictions.

3.4 Proposition 1: bang-bang continuation values

The first major result is that, under interior debt choices and mild regularity, the best equilibrium never uses interior continuation values on path. Instead,

$$V(b(v)) \in \{\underline{V}(b(v)), \bar{V}(b(v))\} \quad \text{on path.}$$

- So the economy is always in either the **best feasible regime** or the **worst feasible regime**.
- These are exactly the two fiscal regimes of the paper.
- The proof uses perturbation arguments: if continuation values were interior, one could compress or spread distortions locally and strictly improve social welfare.

Interpretation. The optimal self-enforcing fiscal system does not use a smooth spectrum of punishments. It uses two extremes: a good regime and a bad regime.

3.5 Proposition 2: the fiscally responsible regime

Define the government's flexible borrowing level under the *best* continuation value as

$$b^r(q, v) \in \arg \max_{b \in [\underline{b}, \bar{b}]} \{\alpha v U(q + b) + \delta \bar{V}(b)\}.$$

In the good regime, the solution to the planner's problem takes the form

$$(b(v), V(b(v))) = \begin{cases} (b^r(q, v), \bar{V}(b^r(q, v))) & \text{if } v < v^*, \\ (b^r(q, v^*), \bar{V}(b^r(q, v^*))) & \text{if } v \in [v^*, v^{**}], \\ (b^p(q, v), \underline{V}(b^p(q, v))) & \text{if } v > v^{**}, \end{cases}$$

where the marginal type v^{**} is indifferent:

$$\alpha v^{**} U(q + b^r(q, v^*)) + \delta \bar{V}(b^r(q, v^*)) = \alpha v^{**} U(q + b^p(q, v^{**})) + \delta \underline{V}(b^p(q, v^{**})).$$

- **Low types** ($v < v^*$). They are unconstrained and borrow flexibly under the best continuation value.
- **Intermediate types** ($v \in [v^*, v^{**}]$). They would like to borrow more, but they stay exactly at the deficit limit to preserve the good regime.
- **High types** ($v > v^{**}$). Their fiscal needs are large enough that they break the deficit limit and accept punishment.

Interpretation. Fiscal responsibility is a *maximally enforced deficit limit*. Reward for restraint is continuation in the good regime; punishment for violating the limit is a transition to the bad regime. This counteracts the political tendency to overborrow.

3.6 Proposition 3: the fiscally irresponsible regime

In the bad regime, the mirror solution is

$$(b(v), V(b(v))) = \begin{cases} (b^r(q, v), \bar{V}(b^r(q, v))) & \text{if } v > v_n^*, \\ (b^r(q, v_n^*), \bar{V}(b^r(q, v_n^*))) & \text{if } v \in [v_n^{**}, v_n^*], \\ (b^p(q, v), \underline{V}(b^p(q, v))) & \text{if } v < v_n^{**}, \end{cases}$$

where the marginal type v_n^{**} is indifferent:

$$\alpha v_n^{**} U(q + b^r(q, v_n^*)) + \delta \bar{V}(b^r(q, v_n^*)) = \alpha v_n^{**} U(q + b^p(q, v_n^{**})) + \delta \underline{V}(b^p(q, v_n^{**})).$$

- **High types** ($v > v_n^*$). They borrow flexibly above the surplus floor and earn a return to the responsible regime.
- **Intermediate types** ($v \in [v_n^{**}, v_n^*]$). They are pushed up to the surplus-limit threshold in order to escape the bad regime.
- **Low types** ($v < v_n^{**}$). They fail to choose enough borrowing, remain in the bad regime, and receive the low continuation value.

Interpretation. Fiscal irresponsibility is a *maximally enforced surplus limit*. This is a punishment: it induces governments to overborrow, sometimes even relative to what they would choose given their own current bias, because returning to the good regime is valuable.

3.7 Why the bad regime is temporary

A key substantive point is that the bad regime is *not* absorbing.

- In proposition 3, the threshold satisfies $v_n^{**} < \bar{v}$.
- Therefore, with strictly positive probability the economy draws a shock high enough that the government respects the surplus limit and returns to fiscal responsibility.
- The paper shows that a temporary punishment is worse for society than an absorbing Markov punishment because it leverages the incumbent's desire to regain the good regime and thereby creates additional overborrowing.

Interpretation. The worst punishment is not permanent chaos. It is a temporary regime that makes current distortion especially severe while still promising future redemption.

3.8 Regime dynamics

Putting propositions 2 and 3 together gives the full law of motion for fiscal regimes.

- **Start in the responsible regime.**
 - If $v \leq v^*$, the government respects the deficit limit and the economy stays in the responsible regime.
 - If $v > v^*$, the government breaks the limit and the economy moves to the irresponsible regime next period.

- **Start in the irresponsible regime.**

- If $v \geq v_n^{**}$, the government chooses borrowing high enough to respect the surplus limit and the economy returns to the responsible regime.
- If $v < v_n^{**}$, the government remains in the irresponsible regime.

Interpretation. Crisis periods trigger switches in *both* directions. In the good regime, crises make the government willing to break discipline. In the bad regime, crises make the government willing to borrow enough to reset the system.

3.9 Analytical example: log utility and saving rates

To study existence and comparative statics, the paper specializes to

$$U(g) = \log g.$$

It then defines a saving rate $s_t \in [0, 1]$ as the fraction of lifetime resources not spent today:

$$g_t = (1 - s_t)R \left(\frac{\tau}{R - 1} - b_{t-1} \right).$$

This parameterization has two important consequences.

- The inherited debt enters welfare additively, which makes continuation equilibria independent of initial debt in terms of future saving-rate choices.
- The gap between best and worst continuation values becomes a constant:

$$\bar{V}(b) - \underline{V}(b) = P^*.$$

So the punishment level does not depend on the inherited debt state.

The responsible and irresponsible thresholds then satisfy

$$\delta P^* = \alpha \int_{v^*}^{v^{**}} [U(g^f(v)) - U(g^f(v^*))] dv,$$

$$\delta P^* = \alpha \int_{v_n^{**}}^{v_n^*} [U(g^f(v_n^*)) - U(g^f(v))] dv,$$

and the punishment itself solves

$$P^* = \delta P^* + \alpha \left[\int_{v^*}^{v^{**}} (U(g^f(v^*)) - U(g^f(v))) Q(v) dv - \int_{v_n^{**}}^{v_n^*} (U(g^f(v_n^*)) - U(g^f(v))) Q(v) dv \right].$$

Here $g^f(v)$ is the government's flexible spending choice under log utility.

Interpretation. These equations summarize the entire regime problem with a single object, P^* , the value of punishment.

3.10 Factorization algorithm and proposition 4

The paper defines an operator $T(P)$: given a future punishment level P , it computes the largest punishment that can be inflicted in the present while respecting incentive constraints.

- $T(P)$ is increasing and concave.
- $T(0) = 0$, so the Markov equilibrium is always a fixed point.
- $\lim_{P \rightarrow \infty} T'(P) < 1$, so there can be at most one positive fixed point.

The main existence result is:

- There exist thresholds $\tilde{\delta} \in (0, 1)$ and $\tilde{\alpha} > 1$ such that if $\delta > \tilde{\delta}$, then

$$\bar{V}(\cdot) > \underline{V}(\cdot) \iff \alpha > \tilde{\alpha}.$$

- If $\delta \leq \tilde{\delta}$, then the unique equilibrium is Markov:

$$\bar{V}(\cdot) = \underline{V}(\cdot).$$

The paper also shows that for an interval of deficit-bias values just above the threshold $\tilde{\alpha}$, regime transitions occur on the equilibrium path.

Interpretation. Two forces are needed for regimes: enough patience to make future punishments matter, and enough present bias to make those punishments credible.

3.11 Numerical simulation

The paper calibrates a simple simulation under

$$\alpha = 1.151, \quad \delta = 0.943, \quad R = 1.05,$$

with v lognormally distributed with mean 0, variance 0.175, and support $[0.01, 100.01]$.

- The deficit-limit threshold spending rate is 0.0736.
- The surplus-limit threshold spending rate is 0.0867.
- Parameter values are chosen so that the mean and variance of the flexible spending rate match the US spending rate over 1970–2020.

Interpretation. In the simulation, the best-equilibrium spending rate is always above the first-best rate because governments are deficit biased. But relative to the flexible Markov benchmark:

- in the responsible regime, the best-equilibrium rate is often *lower* because the deficit limit restrains overspending;
- in the irresponsible regime, the best-equilibrium rate can be *higher* because the surplus limit induces extra borrowing to earn a return to the good regime.

4 Mechanism and assumptions (what makes the result work)

4.1 Why both frictions are necessary

- **Without deficit bias**, there is no need to discipline governments: the first best is self-enforcing.
- **Without private information**, future governments can punish public deviations directly and there is no need for persistent regimes.
- **With both frictions**, the government can deviate privately and future governments cannot condition directly on the shock, so only observed fiscal behavior can be used to create incentives.

4.2 Why the responsible regime is a deficit limit

- The incumbent overvalues current spending, so society wants to cap borrowing when fiscal needs are not too extreme.
- Because future governments share society's preferences from tomorrow onward, continuation in the good regime is a genuine reward.
- The deficit limit is therefore the natural way to offset the overborrowing bias.

4.3 Why the irresponsible regime is a surplus limit

- A punishment must make the present government worse off while remaining incentive compatible.
- Since the incumbent already likes borrowing too much, making it borrow *even more* is the most effective way to destroy welfare.
- A minimum-borrowing requirement (the surplus limit) does exactly this and, combined with the promise of future responsibility, can make governments overborrow even relative to their own flexible choice.

4.4 Why crises trigger regime transitions

- In the responsible regime, a crisis raises the current value of spending enough that the deficit limit is no longer worth respecting.
- In the irresponsible regime, a crisis makes large deficits attractive enough that the government is willing to choose borrowing high enough to escape punishment.
- So crises do not just loosen discipline. They also create the conditions under which discipline can be *reinstated*.

4.5 Why a larger bias makes regimes possible

- If α is close to 1, the planner cannot credibly generate a large enough continuation-value gap between regimes. Concavity makes large punishment gaps too expensive in terms of current distortions.

- As α rises, governments become more willing to choose spending far from first best, so stronger punishments become credible.
- Therefore, the same political force that generates long-run debt pressure also generates the possibility of sustained fiscal discipline through regime threats.

4.6 Relation to the literature

- **Political economy of debt.** Relative to dynamic debt models such as Battaglini–Coate and Yared, this paper adds privately observed fiscal needs and obtains endogenous regime switching.
- **Commitment versus flexibility.** It extends Halac–Yared’s earlier work on fiscal rules by moving from static, exogenously enforced penalties to a dynamic setting where all punishments must be self-enforced.
- **Repeated games.** The bang-bang result resembles the logic of repeated-game punishment regimes, but this paper solves an *adverse-selection* problem with continuous actions and shocks rather than a standard moral-hazard problem.
- **Present bias / hyperbolic discounting.** The paper also contributes to the literature on self-control and time inconsistency by characterizing the welfare-maximizing self-enforcing rule for present-biased policymakers.

5 Figures and proposition map

5.1 Figure 1: empirical motivation

Read:

- Figure 1 plots fiscal consolidations in the United States and the European Union from 1980 to 2014, using the Alesina–Favero–Giavazzi “plan impact” measure.
- The visual point is not just that austerity comes in waves; it is that these waves often begin and end around recessions.
- This is exactly the empirical pattern the model is built to explain: long stretches of fiscal discipline or profligacy, punctuated by crisis-triggered regime shifts.

5.2 Proposition 1 and Figure 2: the two-regime architecture

Read:

- The left panel of figure 2 shows the responsible regime. Debt is increasing in the shock at low values, then flat at the deficit limit, and then jumps to the punished flexible schedule once the limit is broken.
- The right panel shows the irresponsible regime. Debt is flat at the surplus floor for a middle region of shocks and then increases again once shocks are high enough to justify a return to the good regime.

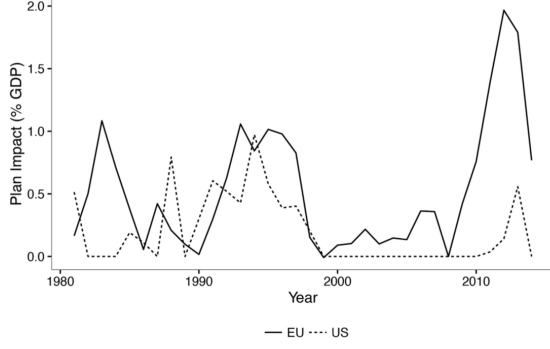


FIG. 1.—Displayed plan impact measure corresponds to exogenous innovations to tax and spending policy, with higher values indicating greater austerity. See the discussion in Alesina, Favero, and Giavazzi (2019) and their figure 6.1.

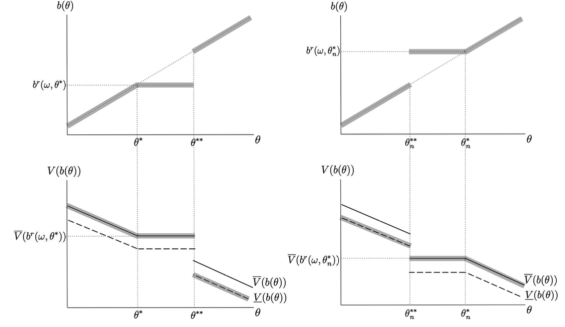


FIG. 2.—Examples of maximally enforced deficit limit (*left*) and maximally enforced surplus limit (*right*). Gray lines depict the prescribed levels of debt in the top panels and the prescribed continuation values in the bottom panels. Solid and dashed black lines in the bottom panels depict $V(b(\theta))$ and $\underline{V}(b(\theta))$, respectively. The figure is drawn under $U(\cdot) = \log(\cdot)$, in which case $V(b(\theta)) - \underline{V}(b(\theta))$ equals a constant independent of $b(\theta)$ and thus $b'(\omega, \theta) = b'(\omega, \theta)$ for all ω and θ (see sec. V).

- The bottom panels show the corresponding continuation values: the economy jumps between $\bar{V}(\cdot)$ and $\underline{V}(\cdot)$ rather than moving smoothly between intermediate values.
- Visually, figure 2 is the cleanest summary of the paper’s message: fiscal policy is organized around two kinked rules, not one smooth reaction function.

5.3 Figure 3: the fixed-point logic of regime existence

Read:

- Figure 3 plots the operator $T(P)$ against the 45-degree line.
- Left panel: if $T'(0) > 1$, the operator crosses the 45-degree line at a positive fixed point $P^* > 0$, so the regime equilibrium exists.
- Right panel: if $T'(0) \leq 1$, the only fixed point is $P = 0$, which corresponds to the Markov outcome with no regimes.
- This figure is the main comparative-statics device of the paper: it explains why patience and deficit bias matter for the existence of fiscal regimes.

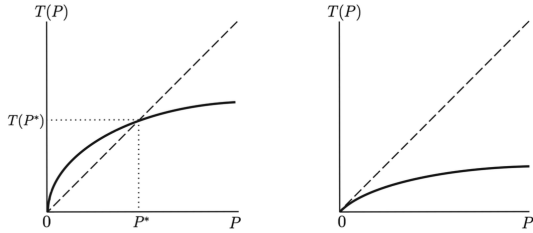


FIG. 3.—Representation of function $T(P)$. The figure depicts scenarios in which $T'(0) > 1$ (*left*) and $T'(0) < 1$ (*right*). Dashed line = 45° line.

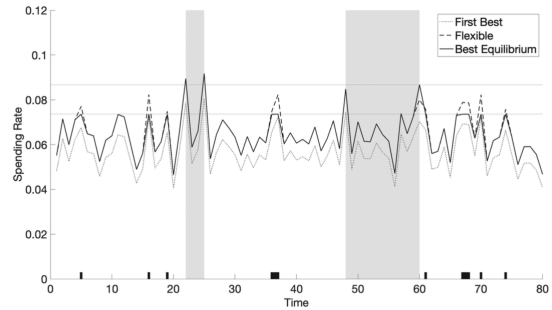


FIG. 4.—We take $U(\cdot) = \log(\cdot)$, $\alpha = 1.151$, $\delta = 0.943$, and $f(\theta)$ lognormal with mean 0, variance $\sigma = 0.175$, and support $[\underline{\theta}, \bar{\theta}] = [0.01, 100.01]$. Bottom horizontal line depicts the flexible spending rate of type θ^* , given by 0.0736; top horizontal line depicts the flexible spending rate of type θ^* , given by 0.0867. Shaded areas correspond to periods of fiscal irresponsibility.

5.4 Figure 4: numerical simulation of regime transitions

Read:

- The shaded regions in figure 4 are periods of fiscal irresponsibility.
- In nonshaded periods, the best-equilibrium spending rate often sits exactly at the deficit-limit threshold when the flexible rate would otherwise be too high.
- At the start of a shaded region, the shock is large enough that the government breaks the deficit limit and moves to the bad regime.
- At the end of a shaded region, the government chooses borrowing high enough to satisfy the surplus limit and restore the good regime.
- The figure makes the regime logic concrete: even with i.i.d. shocks, policy is persistent because today's choices change tomorrow's regime.

5.5 Proposition map: how the results fit together

Read:

- **Lemma 1:** without one of the two frictions, the best equilibrium always restarts and there is no regime logic.
- **Proposition 1:** continuation values are bang-bang, so there are only two regimes on path.
- **Proposition 2:** the good regime is a maximally enforced deficit limit.
- **Proposition 3:** the bad regime is a maximally enforced surplus limit and is temporary.
- **Proposition 4 + corollary:** regimes exist only if patience and deficit bias are high enough, and for a range just above the threshold, transitions occur on path.

6 Short summary

- **Method.** The paper studies the welfare-maximizing self-enforcing equilibrium in an infinite-horizon political-economy model with deficit-biased governments and private fiscal shocks.
- **Main mechanism.** Private information plus limited commitment imply that continuation values become the only disciplining device. The optimal use of those continuation values generates a good regime and a bad regime.
- **Main result.** Fiscal responsibility is a deficit limit; fiscal irresponsibility is a surplus limit used as punishment. Severe shocks can trigger transitions in both directions.
- **Comparative static.** Regimes arise only if governments are sufficiently patient and sufficiently deficit biased. A larger political bias makes future irresponsibility a credible threat, which can support current discipline.

7 Conclusion

7.1 Core conclusion

- The paper provides a theory of why fiscal policy can alternate between persistent responsible and irresponsible regimes even when shocks are i.i.d. and governments are otherwise identical across periods.
- The central ingredient is the interaction of a deficit bias with private information: governments must be disciplined by continuation regimes rather than by direct enforcement.
- The resulting equilibrium is history dependent and crisis driven.

7.2 Economic intuition

- The threat of a bad future can discipline current policymakers only if that bad future is itself credible.
- Credibility comes from the same political distortion that causes debt problems in the first place: future governments are also tempted to overspend.
- Good policy is therefore sustained not by eliminating the bias, but by using the prospect of temporarily worse policy as an incentive device.

7.3 Takeaway

This paper is important because it connects two facts that are often studied separately: long-run political pressure toward debt accumulation and medium-run regime shifts in fiscal policy. Its message is subtle but powerful: the same political economy force that makes governments fiscally irresponsible can also make fiscal responsibility sustainable, provided the threat of future irresponsibility is strong enough. That insight helps explain why fiscal discipline and fiscal crises often arrive together.